

Integrated hybrid energy and time-of-use electricity tariffs for the resource-constrained project scheduling problem

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ABSTRACT

With growing environmental awareness and rising energy costs, projects increasingly face significant challenges in achieving energy efficiency and reducing carbon emissions. To address these issues, this paper integrates hybrid energy, time-of-use (TOU) tariffs and multi-shift scheduling into the resource-constrained project scheduling (RCPS-HE-TOU-MS) framework. A mixed-integer model is formulated to reduce the energy cost and the carbon emissions. During project execution, the model optimizes the proportion of different energy sources and schedules high-energy-consumption tasks during low-electricity-price periods to achieve its objective. Considering the characteristics of the problem, the study introduces a preference-driven multi-objective deep reinforcement learning algorithm that combines exact optimization techniques with an improved multi-step prioritized experience replay dueling double deep Q-network to derive optimal scheduling strategies from training instances. By incorporating spatial pyramid pooling, cosine similarity evaluations and decomposing problems into scalar subproblems, the algorithm enhances adaptability, stability and training efficiency in multi-objective deep reinforcement learning. This model equips project managers with a robust theoretical tool to effectively manage energy consumption and to allocate energy resources rationally, thereby reducing project costs and carbon emissions.

1. Introduction

In order to remain competitive in the global market, companies must continually innovate products and customize production. Consequently, more operational activities are being carried out as projects. Due to the widespread use of project management in high-energy-consumption industries like manufacturing and construction, optimizing energy use, especially efficient electricity scheduling, has become an important research direction in these fields. According to the U.S. Energy Information Administration, global energy demand is projected to increase by 56% between 2010 and 2040 (Sieminski, 2013). Of this, electricity demand makes up around 25% of the total energy supply, and this proportion is steadily rising (EIA, U.S., 2017). This high level of energy consumption leads to significant greenhouse gas emissions, now contributing to over one-third of total carbon dioxide emissions in both developed and developing countries (Min et al., 2022). In order to effectively address climate change and promote sustainable development, incorporating greenhouse gas emissions cost into project cost considerations is crucial. Implementing carbon pricing mechanisms, such as carbon taxes or emission trading systems, enables organizations to assess the environmental impact of their projects and

to promote low-carbon decision making. Estimating greenhouse gas emissions has become an important area of study in project scheduling research. Some studies focus on carbon emissions from electricity consumption, explicitly modeling and analyzing this issue (Aramesh et al., 2023; Hussain and Hussain, 2023; Shishehgharkhaneh et al., 2022). The rapid growth in global electricity consumption exacerbates greenhouse gas emissions and causes electricity shortages and power outages during peak periods. Additionally, the ongoing rise in energy demand places a significant economic strain on the power grid and electricity companies. To address these challenges, the Paris Agreement, effective since November 2016, urges countries to implement demand-side management policies to optimize electricity consumption (Saffari et al., 2018). More countries are now adopting dynamic pricing systems to shift demand away from peak electricity periods. The main types of dynamic pricing systems are real-time pricing (RTP), TOU and critical peak pricing (CPP) (Wesleh Jr. and Lin, 2022). Of these, TOU is the most widely adopted and used form (Ye et al., 2016). Substantial evidence indicates that TOU tariffs effectively reduces average electricity consumption during peak pricing periods (Faruqui and Sergici, 2010; Faruqui et al., 2017; Dutta and Mitra, 2017). For instance, in September

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2022, an intense heatwave in California pushed the market load managed by the California Independent System Operator (CAISO) to record levels. To prevent widespread rolling blackouts, an emergency text alert was sent to millions of residents, urging them to “conserve energy immediately to protect public health and safety.” According to the CAISO president, this message reduced system load by approximately 4%, likely preventing costly and large-scale grid disruptions.

The growing demand for sustainable development has garnered significant attention from both academia and industry, particularly towards alternative energy sources, with a focus on renewable energy. In recent years, renewable energy development and utilization have grown substantially, accounting for over half of the global increase in electricity generation (Mondal et al., 2023). The solar photovoltaic (PV) industry has become a key component of the global renewable energy system. The development of microgrid technology has encouraged more companies to deploy distributed solar power systems, promoting self-consumption, reducing electricity costs and decreasing environmental impact. Many companies have adopted solar rooftop installations to meet the electricity demands of their production facilities. In 2019, Alois Müller GmbH in Germany successfully used a solar rooftop to supply two-thirds of the factory’s carbon-neutral electricity (e-con A.G., 2024). Solar rooftop technology has advanced to the point where it can now meet up to 25% of a factory’s peak energy demand Yang (2023). Leading companies, including BMW, BOE Technology Group and Linuo Ritter in Mainland China, as well as Taiwan’s Beitai Broadcast City, have actively explored and implemented solar rooftop technology. Many projects, such as aircraft assembly and shipbuilding, require large factory spaces for assembly. Installing solar power equipment on large rooftops not only converts solar energy into electricity for factory needs but also provides shading and thermal insulation, reduces air conditioning costs, and extends the roof’s lifespan.

To ensure the stability and reliability of the energy supply during project execution, a combination of solar power (green electricity, GE) and power grid can be employed. By integrating GE with traditional energy sources and utilizing energy microgrids, smart energy systems and energy internet technologies, the environmental sustainability, economic viability and flexibility of projects can be substantially improved. To optimize the utilization and productivity of critical equipment, manufacturing and construction industries typically implement multi-shift workdays, often dividing them into day and night shifts. TOU tariffs allow the calculation of average prices for each shift, which tend to fluctuate. Consequently, project managers can schedule energy-intensive tasks during shifts with lower electricity prices to minimize project costs.

Currently, systematic and scientific decision-making methods are lacking to address RCPSPs that simultaneously consider multi-shift systems, renewable energy and grid power supply, and TOU tariffs. These methods would help project managers to formulate optimized scheduling plans under precedence and resource relationship constraints. This paper proposes an RCPSP that integrates dual-energy hybrid power supply, TOU tariffs and multi-shift systems. In this study, the renewable resources consist primarily of high-energy-consuming critical equipment capable of continuous operation. Activities involving high-energy-consuming critical equipment should be scheduled during periods of lower electricity prices. In multi-shift systems, activities are quantified in shifts, and any activity can be preempted. Activities can be interrupted in one shift and resumed in a later shift. Since labor costs remain relatively stable compared to energy costs under TOU tariffs, they are neglected. The RCPSP-HE-TOU-MS is a bi-objective project scheduling problem with goals to minimize energy-related costs (including renewable energy, power grid and carbon quota costs) and carbon emissions, subject to precedence and resource constraints.

The RCPSP is known to be an NP-hard problem (Blazewicz et al., 1983), and its extended version, RCPSP-HE-TOU-MS, clearly retains this NP-hardness. Exact algorithms are often impractical for large-scale problems within reasonable time limits, necessitating efficient

optimization methods to obtain optimal solutions in a feasible time. The rapid advancement of artificial intelligence has led to the widespread application of deep reinforcement learning (DRL) in combinatorial optimization problems, owing to its remarkable learning capabilities (Choi and Han, 2023; Kedir et al., 2022). DRL enables the accumulation of experience during training, which can be applied to solving new problems. A preference-driven multi-objective deep reinforcement learning (PD-MODRL) algorithm is proposed based on DRL theory and the characteristics of the problem. The algorithm is initially trained on mixed instances with varying numbers of activities. By diversifying the training instances, the model learns a broader range of features, improving its generalization ability. Additionally, the algorithm incorporates spatial pyramid pooling techniques and uses cosine similarity evaluations to align with preset preferences, thereby significantly improving the adaptability and stability of multi-objective deep reinforcement learning in dynamic environments. Moreover, by leveraging a problem decomposition approach, the proposed method disaggregates the multi-objective optimization task into scalar subproblems and employs neighborhood parameter transfer alongside reinforcement learning to collaboratively optimize each sub-model, thereby substantially enhancing training efficiency.

The following three aspects reflect the main contributions of this paper:

- (1) A new variant of the RCPSP is proposed that integrates renewable energy, TOU tariffs and multi-shifts into its framework. Renewable energy and TOU tariffs complement each other and are key factors influencing energy-related costs and carbon emissions in energy-intensive projects. This study incorporates these two factors into the traditional RCPSP, reducing energy-related costs and carbon emissions. This enhances the flexibility of project managers in managing energy-intensive projects.
- (2) A bi-objective mixed-integer linear programming model for the RCPSP-HE-TOU-MS has been developed. The model integrates two coupled aspects, project scheduling and electricity allocation, based on the characteristics of high-energy-consuming projects. The constraints of the RCPSP are extended to account for these considerations.
- (3) An intelligent algorithm based on deep reinforcement learning was designed to address the specific characteristics of the problem. Experimental data confirmed that this framework achieves high problem-solving accuracy while maintaining algorithmic efficiency.

The structure of the paper is as follows. Section 2 reviews the relevant literature. Section 3 introduces the RCPSP-HE-TOU-MS and associated models. The PD-MODRL algorithm is presented in Section 4. Section 5 evaluates the performance of the proposed algorithms through computational experiments. A case study illustrating the problem is discussed in Section 6. Finally, Section 7 concludes the paper and outlines directions for future research.

2. Literature review

This study extends the classic RCPSP by incorporating preemptive scheduling, which enables flexible activity splitting to improve resource utilization. Furthermore, it integrates energy-aware scheduling by considering energy consumption in project planning to enhance sustainability. In addition, the paper explores the application of deep reinforcement learning techniques, emphasizing their potential to improve adaptive decision-making in project scheduling.

2.1. Preemptive RCPSP

In the preemptive RCPSP (PRCPSP), activities can be split into multiple discrete segments at predefined time points. This strategy,

Table 1
Preemptive RCPSP literature.

Publication	Objective	Renewable energy	TOU	Solution method	Comment	Year
Demeulemeester and Herroelen	f_n			B&B	Innovatively applies B&B to PRCPSP	1996
Saeidi et al.	C_{max}, R			TH		2020
Artigues et al.	C_{max}			constr.	First study on MSPSP-PP with partial preemption	2021
Mirnezami et al.	C_{max}, Z_2, Z_3			AUGMECON2		2023
Peng et al.	D, C		✓	ϵ -constraint method, GA	Innovatively integrates TOU tariffs with project scheduling	2025

* f_n : project duration; B&B: branch and bound; C_{max} : project completion time; R : total manpower available; Z_2 : total resource cost; Z_3 : total project risk; D : project duration; C : energy cost; TH: horabi-hosseini; constr.: constructive algorithm; AUGMECON2: augmented epsilon-constraint method; GA: genetic algorithm.

known as activity splitting, enhances resource allocation efficiency and improves project objectives, as established by Demeulemeester and Herroelen (1996). Saeidi et al. (2020) present a PRCPSP model that optimizes labor utilization and project completion time by permitting activities to be interrupted and resumed. Artigues et al. (2021) proposed multiple mathematical models to minimize the project completion time through resource-constrained scheduling, specifically addressing preemptive task execution and resource skill heterogeneity.

Mirnezami et al. (2023) developed a multi-objective mixed integer linear programming model to address the PRCPSP under multi-skilled and multi-mode conditions. The model simultaneously optimizes three objectives: minimizing project duration, reducing total resource costs and mitigating total project risk. Notably, the study incorporates uncertainty through a lognormally distributed parameter representing the average non-renewable resource demand. Peng et al. (2025) investigated a MRCPSP-MS-TOU. They formulated a bi-objective MILP model and solved it by the ϵ -constraint method and meta-heuristics. A notable feature of their study is the consideration of preemptive scheduling, where energy-intensive activities requiring critical equipment can be interrupted and shifted to subsequent low-price shifts, thus reducing costs and project duration under TOU tariffs. The PRCPSP introduces notable computational complexity due to the combinatorial nature of preemptive scheduling combined with multi-dimensional optimization objectives. This complexity has driven recent research toward heuristic approaches for large-scale problem instances. Established meta-heuristic algorithms, including genetic algorithms (Delgoshaei et al., 2019; Du et al., 2021) and ant colony optimization (Magsoudlou et al., 2019), have demonstrated effectiveness in addressing PRCPSP challenges, particularly in balancing solution quality with computational efficiency. To summarize prior research on preemptive scheduling approaches and their relevance to RCPSP, the key studies are compared in Table 1.

In this paper, the RCPSP-HE-TOU-MS is classified as the PRCPSP defined by Cheng et al. (2015), which allows the activity to be split at any discrete time instant. In contrast to previous research, the activities in RCPSP-HE-TOU-MS proposed in this paper are divided by shifts instead of by hours or days.

2.2. Energy-aware scheduling and project scheduling problem with energy awareness

Energy-aware scheduling has received significant attention in recent years, with many scholars investigating its potential to optimize energy utilization and to improve the efficiency of energy systems. However, existing research primarily focuses on traditional thermal

power sources, especially fossil fuels like coal and oil, while giving limited attention to clean energy and dynamic electricity pricing factors (Paraschiv et al., 2014; Sun et al., 2020). To promote energy conservation and to reduce emissions, many countries have set renewable energy development targets and implemented dynamic electricity pricing mechanisms (Nie et al., 2018). To tackle the intermittency and uncertainty of renewable energy, Ji et al. (2019) and Wang (2018) proposed the concept of multi-energy complementary systems. This system integrates various clean energy sources, along with electricity, cooling and heating, effectively reducing environmental pollution from traditional energy sources while ensuring the stability of renewable energy supply. In engineering and manufacturing projects, the use of multi-energy complementary systems has gradually expanded to various industries. For instance, Yuan and Dornfeld (2009) assessed the environmental and economic benefits of three alternative energy sources compared to conventional electricity in the automotive manufacturing industry. Similarly, Zhao et al. (2011) examined the use of hybrid energy complementary power supplies in military equipment.

Research in workshop scheduling that incorporates renewable energy supply and TOU tariffs remains scarce. Wu et al. (2020) proposed a mathematical model for the flexible flow shop scheduling problem, focusing on minimizing both makespan and carbon emissions. They solved this model using a fast non-dominated sorting genetic algorithm combined with a low-carbon scheduling strategy. Ghorbanzadeh and Ranjbar (2023) investigated the energy-aware flow shop scheduling problem, considering sequence-dependent setup times, group scheduling and renewable energy constraints. They formulated two mixed-integer linear programming models and proposed a decomposition-based heuristic algorithm to efficiently solve medium-sized instances. Dong et al. (2022) addressed the reentrant hybrid flow shop scheduling problem by proposing a two-stage scheduling optimization model that incorporates renewable energy. They designed a hybrid algorithm combining the salp swarm algorithm with NSGA-II, demonstrating its feasibility in low-carbon manufacturing. Gong et al. (2019) considered TOU tariffs, labor costs and machine switching in the job shop scheduling problem. They constructed a five-objective model and algorithm based on NSGA-II. Kurniawan et al. (2021) addressed a bi-objective job shop scheduling problem with TOU tariffs, focusing on minimizing both delay and total energy costs. They solved the problem using a distributed elite local search algorithm based on genetic algorithms. Shen et al. (2023) proposed a job shop scheduling problem under TOU tariffs, where the makespan is treated as a constraint, and the objective is to minimize total energy costs. They developed a heuristic method based on iterative tabu search algorithms.

The application of energy awareness in project scheduling remains limited. [Zheng and Wang \(2015\)](#) proposed a bi-objective RCPSP that optimizes both the project duration and the energy consumption. The model integrates the duration and energy consumption of each activity, where shortening the duration results in an increased energy consumption. It formulates the problem using bi-objective mixed-integer programming. To solve the problem, a Pareto-based estimation of distribution algorithm is proposed. [Okubo et al. \(2015\)](#) investigated the use of partially renewable resources within the RCPSP framework, accounting for energy consumption during peak periods and power constraints. They proposed a constraint-based planning approach, integrated with heuristic-pattern-restricted masking algorithms. [Magsoudlou et al. \(2021\)](#) examined the use of multi-skilled labor in preemptive project scheduling, focusing on the substantial rise in energy prices during peak periods. They modeled the minimization of the total energy consumption under TOU tariffs. [Du et al. \(2021\)](#) tackled project scheduling with multi-granularity tasks, proposing the splitting and recombining of activities into tasks of varying sizes. They developed a bi-objective scheduling model to minimize project delay and total energy costs under TOU tariffs, proposing an enhanced NSGA-II algorithm to solve the problem. [Javanmard et al. \(2022\)](#) examined energy efficiency with TOU tariffs in multi-project, multi-skilled scheduling. They proposed a bi-objective RCPSP model to minimize the total cost of multi-skilled labor and project priority deviation, and developed an intelligent multi-objective evolutionary algorithm integrating cuckoo search, particle swarm optimization and genetic algorithms. [Karnebogen and Zimmermann \(2024\)](#) made significant progress in addressing the RCPSP/max-X by integrating partially renewable resources with generalized precedence constraints. They proposed two efficient solution-generation schemes: a construction-based heuristic and a relaxation-based approach. These schemes can rapidly produce feasible solutions for most benchmark instances while substantially improving project duration and resource utilization, providing new and effective strategies for incorporating renewable energy into project scheduling. [Chen et al. \(2025\)](#) integrated renewable energy into project scheduling, demonstrating that coordinated optimization of energy resources and task timing under TOU tariffs can reduce costs and improve renewable utilization. [Table 2](#) provides an overview of existing studies on energy-aware project scheduling.

2.3. Reinforcement learning in project scheduling problems

With the advancement of artificial intelligence technologies, researchers have increasingly explored the application of reinforcement learning (RL) to combinatorial optimization problems, such as the vehicle routing problem ([Basso et al., 2022](#)), the traveling salesman problem ([Bogrybayeva et al., 2023](#)) and three-dimensional bin packing problem ([Zhao et al., 2022](#)). Despite these advancements, the application of RL in project scheduling problems remains relatively underexplored. Some studies have employed an approximate dynamic programming (ADP) approach based on the Rollout policy to address project scheduling under uncertainty. The fundamental concept behind this ADP approach is analogous to tabular RL algorithms, and it was first introduced by [Li and Womer \(2015\)](#) as a method for solving stochastic resource-constrained project scheduling problems (SRCPPs). [Choi et al. \(2007\)](#) were the first to introduce reinforcement learning techniques into the dynamic project arrival variant of the SRCPP. The scheduling process was formulated as a Markov Decision Process (MDP), and Q-learning was applied to eliminate irrelevant transition rules that are difficult to incorporate into heuristic algorithms. Subsequent studies have extended this methodology to various other project scheduling challenges. For instance, [Zhao and Xu \(2021\)](#) applied it to the stochastic decentralized multi-project scheduling problem with resource transfers, while ([Xie et al., 2021](#)) utilized it in the context of the resource-constrained project scheduling problem with uncertain resource availabilities. Additionally, [Li et al. \(2023\)](#) adapted

the approach to address dynamic resource leveling in the presence of multiple uncertainties.

Several studies have incorporated RL within their algorithms to address project scheduling problems, with a notable trend toward developing RL-based metaheuristic approaches. Typically, tabular RL algorithms are employed as high-level strategies to guide the selection of low-level metaheuristic algorithms for population updates. For example, [Jędrzejowicz and Ratajczak-Ropel \(2014\)](#) applied an A-team framework as an RL strategy to solve the RCPSP. Under the RL framework, individuals from the population were directed to one of four distinct metaheuristic algorithms for further updates. A similar A-team framework was also used to tackle the MRCPPSP ([Jędrzejowicz and Ratajczak-Ropel, 2015](#)). Additionally, [Sallam et al. \(2021\)](#) proposed a Q-learning-based hyper-heuristic algorithm for solving the SRCPPSP.

In their approach, Q-learning functioned as the upper-level strategy, while the lower-level heuristics included the multi-operator differential evolution algorithm and the discrete cuckoo search algorithm. The population's state, represented by its diversity and solution quality, guided the upper-level algorithm to select an appropriate heuristic for updating the population. In another study, [Shen et al. \(2018\)](#) introduced a Q-learning-based memetic algorithm to address the multi-objective dynamic software project scheduling problem. In this case, two continuous indicators were used to evaluate the current state, and Q-learning was utilized to choose from four actions that combined two local and two global search algorithms. The change in the hypervolume indicator served as the reward function. [Table 3](#) compares representative applications of reinforcement learning in project scheduling.

2.4. Research gaps

A systematic examination of current research reveals significant knowledge gaps in sustainable project scheduling methodologies, particularly concerning energy-conscious planning and emission mitigation strategies. Three critical research deficiencies emerge from the comprehensive literature analysis: Firstly, existing studies predominantly concentrate on conventional single-energy models for the RCPSP. However, innovative approaches integrating renewable energy sources with TOU tariffs have yet to be systematically explored in current scheduling frameworks. Secondly, the academic community has largely overlooked the development of comprehensive energy-aware scheduling systems. The current literature demonstrates a limited investigation into hybrid scheduling models that synergistically combine shift-based management, renewable energy utilization and dynamic electricity pricing mechanisms within project scheduling paradigms. Thirdly, methodological limitations persist in optimization techniques. While conventional heuristic algorithms continue to dominate solution approaches, the application of advanced machine learning techniques, particularly reinforcement learning architectures, for addressing multi-dimensional RCPSP challenges remains insufficiently developed in contemporary research.

3. Problem description and mathematical model

3.1. Problem description and assumptions

In traditional project execution, the energy supply usually relies on a single source, primarily conventional fossil fuels. This study, aiming to achieve sustainable development goals, explores integrating GE with power grid to meet the increasing energy demands during project execution. GE is a promising renewable source but is highly uncertain and intermittent, which can undermine the stability and reliability of the power supply if used exclusively. To address these challenges, microgrid technology is commonly used to improve the stability and reliability of GE. Microgrid technology integrates distributed energy resources, storage systems and intelligent controls to improve

Table 2
Preemptive RCPSP literature.

Publication	Objective	Renewable energy	TOU	Solution method	Comment	Year
Wu et al.	f_1, f_2	✓		GA	Innovatively integrates renewable energy into scheduling optimization	2020
Ghorbanzadeh and Ranjbar	TEC^a	✓	✓	DBHA	Using renewable energy and TOU tariffs in flow shop scheduling	2023
Dong et al.	C_{max}, TC, ZC	✓	✓	IMOWSA	Considering multi-energy complementary	2022
Gong et al.	$C_{max}, TEC^b, TLC, MWL, TWL$		✓	NSGA-III	Considering both energy and labor awareness under dynamic electricity pricing	2019
Kurniawan et al.	F_1, F_2		✓	DEGA	Innovatively applies TOU tariffs policy in job-shop scheduling	2021
Shen et al.	TEC^b		✓	ITS	Considering TOU tariffs	2023
Zheng and Wang	C_{max}, CE_{max}			PBEDA	First time using project scheduling model to address low-carbon production scheduling problem	2015
Okubo et al.	C_{max}	✓		constr.	Proposing a new variant of RCPSP to deal with realistic energy constraints	2015
Maghsoudlou et al.	TEC^a		✓	EMA	First work in project scheduling considering TOU tariffs for energy price to schedule preemptable activities within a multi-skilled environment	2021
Du et al.	Z_1, Z_2		✓	NSGA-II		2021
Javanmard et al.	G_1, G_2		✓	FSCS	Integrating multi-project scheduling with multi-skilled workforce and TOU energy tariffs	2022
Karnebogen and Zimmermann	C_{max}	✓		constr.	Integrating partially renewable resources	2024
Chen et al.	TEC^b	✓	✓	MD3QN-PER		2025

* f_1 : project duration; f_2 : carbon emission; TEC^a : total electricity cost; C_{max} : interval makespan; TC : total interval carbon emissions; ZC : total interval cost; TEC^b : total energy cost; TLC : total labor cost; MWL : maximal machine workload; TWL : total machine workload; F_1 : total weighted tardiness; F_2 : electricity cost; CE_{max} : carbon emissions; Z_1 : project delay; Z_2 : energy consumption cost; G_1 : the total cost; G_2 : the average absolute difference between the projects' priorities in the planning frames and the percentage of the projects' achieved progress; GA: genetic algorithm; DBHA: decomposition-based heuristic algorithm; IMOWSA: improved multi-objective whale swarm algorithm; NSGA-III: nondominated sorting genetic algorithm-III; DEGA: distributed-elite local search based on GA; ITS: iterative tabu search; PBEDA: pareto-based estimation of distribution algorithm; constr.: constructive algorithm; EMA: electromagnetic like algorithm; FSCS: fuzzy dominance sorting cuckoo search; MD3QN-PER: multi-step dueling double deep Q-network with prioritized experience replay.

the energy efficiency, the renewable energy use and the power system reliability and sustainability. To predict the GE availability, the project uses environmental databases and Internet of Things-based prediction methods to estimate the GE production for each time period. During each interval, the PV system directly supplies part of the project's electricity demand, with any surplus stored in the energy storage system. The storage system has a capacity limit, and any excess electricity is sold to the grid company. Additionally, TOU tariffs cause fluctuations in the cost of purchasing electricity from the grid, varying across different time intervals. Based on these considerations, the RCPSP-HE-TOU-MS model is proposed, as detailed below.

The RCPSP-HE-TOU-MS can be modeled using a directed acyclic graph $G = (V, E)$, where $V = 1, 2, \dots, J$ denotes the nodes (activities) with nodes 1 and J as the starting and ending dummy activities, respectively. The edge set EP indicates precedence constraints; if $(i, j) \in E$, then activity j can only begin after activity i has been completed. To indicate the start and end of a project, activities $j = 1$ and $j = J$ are defined as dummy activities. These activities do not require any resources or time. Durations of activities are measured in shifts, with each activity j capable of being split d_j times. A workday comprises several shifts, each with a distinct average electricity price. Although electricity prices generally vary hourly under TOU tariffs, the price for

Table 3
Preemptive RCPSP literature.

Publication	Objective	Solution method	Comment	Year
Li and Womer	C_{max}	ADP	Tractable closed-loop algorithm for SRCPSP	2015
Choi et al.	ER	Q-Learning	Innovatively applies Q-Learning to handle dynamic project arrivals and complex state transitions in stochastic RCPSP	2007
Zhao and Xu	ER	Rollout-based ADP	Proposing a new rollout policy-based ADP algorithm for solving the stochastic decentralized multi-project scheduling problem	2021
Xie et al.	C_{max}	ADP	First closed-loop ADP algorithm for SRCPSP-URA with sequential improvement property	2021
Li et al.	f	HOC-ADP	Combining multiple uncertainties with dynamic updates in resource balancing for the first time	2023
Jedrzejowicz and Ratajczak-Ropel	C_{max}	RL	Proposing a dynamic learning strategy using reinforcement learning	2014
Jedrzejowicz and Ratajczak-Ropel	C_{max}	A-Team with RL	Proposing a dynamic learning strategy for A-Team using RL to efficiently solve MRCPSP	2015
Sallam et al.	C_{max}	DECSwRL	Proposing a multi-method approach combining RL with DE and CS	2021
Shen et al.	f_1, f_2, f_3, f_4, f_5	MOTAMAQ	Proposing a more realistic MODSPSP model	2018

* C_{max} : project completion time; ER : total expected reward; f : total cost of resource usage; f_1 : project duration; f_2 : project cost; f_3 : schedule robustness; f_4 : stability of the project; f_5 : employees satisfaction; ADP: approximate dynamic programming; HOC-ADP: hybrid open closed-loop approximate dynamic programming; A-Team with RL: asynchronous team with reinforcement learning; DECSwRL: differential evolution and cuckoo search with reinforcement learning; MOTAMAQ: multi-objective two-archive memetic algorithm based on Q-learning.

each shift is based on its average cost. Consequently, the unit energy cost on shift ω , fluctuates throughout the workday shifts due to the TOU tariff. The RCPSP-HE-TOU-MS seeks to minimize both energy costs and carbon emissions. The energy cost includes expenses for GE, power grid electricity, and carbon emission allowances. This model operates within a multi-shift system, where energy costs fluctuate between shifts due to the TOU tariffs. By scheduling segments during shifts with lower electricity prices, energy costs can be reduced.

In conclusion, project managers overseeing energy-intensive projects that involve shift systems, renewable energy, and TOU tariffs face several critical decision-making challenges. First, they must decide whether to consistently prioritize renewable energy to reduce carbon emissions, even at the cost of potentially higher energy expenses. Second, they need to balance the scheduling of high-energy tasks during periods of lower electricity rates to minimize energy costs, though this may result in increased carbon emissions. Therefore, minimizing energy costs necessitates the development of a scheduling strategy that aligns high-energy activities with low-cost electricity periods. At the same time, the efficient allocation of renewable energy is crucial to reducing carbon emissions. To date, no study has integrated TOU tariffs, renewable energy and multi-shift systems within the context of the RCPSP, particularly for energy-intensive projects.

3.2. Mathematical model

This model focuses on optimizing the scheduling of start times for each project activity shift while determining the proportions of GE and power grid to be used at different times, with the goal of minimizing overall energy consumption costs and carbon emissions. Additionally, to prevent significant increases in electricity prices due to peak power loads surpassing a predetermined threshold, it is crucial that the activity schedule ensures the total power demand at any given time remains below this limit.

The assumptions involved to formulate the RCPSP-HE-TOU-MS are as follows:

- Renewable resources are mainly composed of critical equipment with a high energy consumption, which is capable of continuous operation.
- The availability of GE is constrained, and its amount is predetermined for each time period.
- The carbon emission per unit of electricity consumed from the grid is constant.
- The peak power load thresholds are adhered to throughout the project's execution.
- Each activity's execution time is a positive integer.

Due to the influence of renewable energy availability and TOU tariffs, traditional project scheduling approaches that aim for non-delay solutions may no longer be optimal. In some cases, delaying the start time of specific activities can lead to a reduction in total costs. Assume the project contract duration is 5, with other data such as electricity pricing and carbon quotas obtained using the same methods as in Section 5.1. To understand the proposed RCPSP-HE-TOU-MS, we present an example of a construction project comprising four real activities, as shown in Fig. 1(a). Activity 1 and activity 6 are dummy activities. Comparison of Fig. 1(b) and Fig. 1 (c) shows that within the contractual period, delaying the start time of activities, considering TOU tariffs and renewable energy, effectively reduces the total costs.

The parameters and variables involved to formulate RCPSP-HE-TOU-MS are as follows:

Index and sets	
j	Index of activity, $j = 1, 2, \dots, J$;
t	Index of time, $t = 1, 2, \dots, T$ in shifts;
$P(j)$	Set of precedence leading into activity j ;
D	Duration of contract;
W	Set of shifts of a workday, $\omega \in W$;
Parameters	
T	Upper bound on the project makespan;

d_j	The duration of activity j , which is the number of shifts required to complete the activity;
EF_j	Early finish time of activity j in shifts;
LF_j	Latest finish time of activity j in shifts;
P_{max}	The threshold of peak power load;
P_j	The power consumption of the equipment during the execution of activity j ;
Q	The capacity of energy storage systems;
Z	Carbon emission allowances;
C_s	The cost of GE per unit;
$C_{f\omega}$	The unit price of electricity purchased from the grid on shift ω ;
C_e	The unit price for purchasing/selling additional carbon emission allowances;
C_ξ	The price of selling GE;
μ	The carbon emission coefficient of electricity from the unit power grid;
Decision variables	
$x_{jet\omega}$	$x_{jet\omega} = 1$ indicates that the segment e of activity j is performed on shift ω of workday t , for $j \in V$, $e \in 1, 2, \dots, d_j$, $t \in 1, 2, \dots, T$, $\omega \in W$; and $x_{jet\omega} = 0$ otherwise;
Other variables	
$P_{t\omega}^s$	The amount of GE available during a unit period t on shift ω ;
$Q_{t\omega}^b$	The remaining energy capacity of the energy storage system at the beginning of period t on shift ω ;
$Q_{t\omega}^d$	The discharge amount of the energy storage system during period t on shift ω ;
$Q_{t\omega}^c$	The amount of energy stored in the energy storage system by the end of period t on shift ω ;
$Q_{t\omega}^f$	The amount of grid electricity consumed during period t on shift ω ;
$Q_{t\omega}^s$	The amount of GE consumed during period t on shift ω ;
$Q_{t\omega}^\xi$	The amount of electricity fed into the grid during period t on shift ω ;
EC	Energy costs;
CEC	Carbon emission costs;
CE	Carbon emissions.

3.2.1. Objective function

The EC include expenses for GE and electricity purchased from the grid.

$$EC = \sum_{t=1}^T \sum_{\omega \in W} \left(C_s \cdot Q_{t\omega}^s + C_{f\omega} \cdot Q_{t\omega}^f - C_\xi \cdot Q_{t\omega}^\xi \right) \quad (1)$$

The process of acquiring energy from the grid contributes to carbon emissions. The remaining carbon emission allowances can be sold externally, and in cases of insufficiency, additional carbon emission allowances must be purchased.

$$CEC = C_e \cdot \left(\sum_{t=1}^T \sum_{\omega \in W} \mu \cdot Q_{t\omega}^f - Z \right) \quad (2)$$

Carbon emissions can be calculated according to Eq. (3).

$$CE = \sum_{t=1}^T \sum_{\omega \in W} \mu \cdot Q_{t\omega}^f \quad (3)$$

The objective function of the model is to minimize the total cost (TC) and CE.

$$\text{Min } TC = EC + CEC \quad (4)$$

$$\text{Min } CE \quad (5)$$

3.2.2. Constraints

In the optimization of project scheduling, it is essential to account for both the internal precedence between activities and the constraints related to energy supply and consumption. Constraints (6)–(12) specifically define the limitations on internal precedence and the peak power limits during the project execution phase.

$$\sum_{t=EF_j}^{LF_j} \sum_{\omega \in W} x_{jet\omega} = 1, \forall j \in V, e = 1, 2, \dots, d_j \quad (6)$$

$$\sum_{t=EF_j}^{LF_j} \sum_{\omega \in W} \sum_{e=1}^{d_j} x_{jet\omega} = d_j, \forall j \in V \quad (7)$$

$$\sum_{e=1}^{d_j} x_{jet\omega} \leq 1, \forall j \in V, t = 1, 2, \dots, T, \omega \in W \quad (8)$$

$$\sum_{t=EF_j}^{LF_j} \sum_{\omega \in W} ((t-1) \cdot |W| + \omega) \cdot x_{jet\omega} < \sum_{t=EF_j}^{LF_j} \sum_{\omega \in W} ((t-1) \cdot |W| + \omega) \cdot x_{j(e+1)t\omega}, \forall j \in V, e = 1, 2, \dots, d_j - 1 \quad (9)$$

$$\sum_{t=EF_i}^{LF_i} \sum_{\omega \in W} ((t-1) \cdot |W| + \omega) \cdot x_{id_t\omega} < \sum_{t=EF_j}^{LF_j} \sum_{\omega \in W} ((t-1) \cdot |W| + \omega) \cdot x_{j1t\omega}, \forall (i, j) \in E \quad (10)$$

$$\sum_{\omega \in W} \sum_{t=EF_j}^{LF_j} ((t-1) \cdot |W| + \omega) \cdot x_{j0t\omega} \leq D \quad (11)$$

$$\sum_{j=1}^J \sum_{e=1}^{d_j} \sum_{t=\text{Max}\{t, EF_j\}}^{\text{Min}\{t+d_j-1, LF_j\}} x_{jet\omega} \cdot P_j \leq P_{max}, t = 1, 2, \dots, T, \forall \omega \in W \quad (12)$$

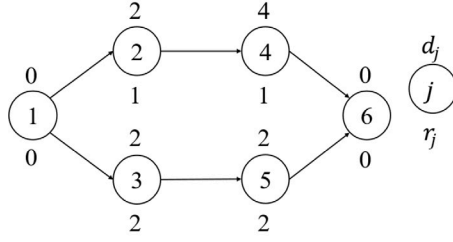
Constraints (6) ensure that each active segment is executed only once. Constraints (7) indicate that the duration of each activity is equal to its number of segments. Constraints (8) ensure that each activity can only be executed during one segment per shift, meaning that no more than one segment of an activity can be performed during the same shift within a workday. Constraints (9) indicate the precedence constraints between adjacent segments of an activity, where the $(e+1)$ -th segment must follow the completion of the e th segment. Constraints (10) define the precedence between activities. Specifically, for every pair of activities $(i, j) \in E$, the initiation of the first segment of activity j is contingent upon the completion of the final segment of activity i . The index d_i denotes the final segment of activity i . Consequently, the decision variable $x_{id_t\omega} = 1$ indicates that the final segment of activity i is scheduled during the ω -th shift on the t th working day. Constraints (11) ensure that the project is completed within the contractual duration. Constraints (12) impose a limit on the maximum peak power. In practical projects, this constraint can be met by adjusting the scheduling sequence or postponing specific activities. For example, if the calculated peak power exceeds the limit, activities can be rescheduled or postponed to keep total power within the allowable threshold.

Constraints (13)–(16) describe the interactions and dependencies related to the utilization of renewable energy:

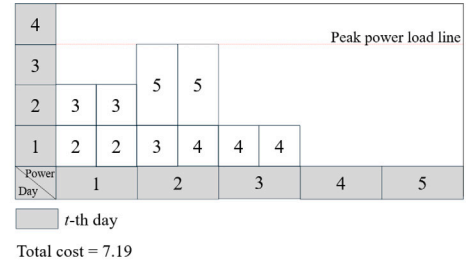
$$Q_{t\omega}^s + Q_{t\omega}^d + Q_{t\omega}^f = \sum_{j=1}^J \sum_{e=1}^{d_j} \sum_{t=\text{Max}\{t, EF_j\}}^{\text{Min}\{t+d_j-1, LF_j\}} x_{jet\omega} \cdot P_j, t = 1, 2, \dots, T, \omega \in W \quad (13)$$

$$Q_{t\omega}^s \leq Q_{t\omega}^b + P_{t\omega}^s, t = 1, 2, \dots, T, \omega \in W \quad (14)$$

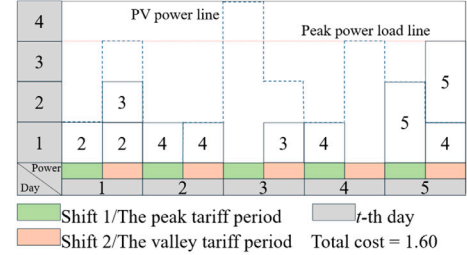
$$Q_{t\omega}^b = \begin{cases} 0, & t = 1, 2, \dots, T, \omega \in W \\ \min \{ Q_{t-1,\omega}^b - Q_{t-1,\omega}^d + Q_{t-1,\omega}^c, Q \}, & t = 2, \dots, T, \omega \in W \end{cases} \quad (15)$$



(a) Example of an activity network



(b) Scheduling without considering TOU tariffs and hybrid energy



(c) Scheduling with considering TOU tariffs and hybrid energy

Fig. 1. An illustrative example for considering TOU tariffs and hybrid energy.

$$x_{jetow} = \{0, 1\}, \forall j \in V, e = 1, 2, \dots, d_j, t = 1, 2, \dots, T, \omega \in W \quad (16)$$

The energy demand of the project is jointly satisfied by GE and power grid electricity, as explicitly defined in Constraints (13). Constraints (14) ensure that the discharging capacity of the microgrid does not exceed the sum of its stored energy and the electricity generated in the current period. Constraints (15) specify that during the first shift the energy storage system does not store any renewable energy, reflecting the fact that the system starts from an empty state at the beginning of the scheduling horizon. In subsequent shifts, the usable energy is the sum of the remaining energy from the previous shift and the stored energy of the current shift, with the total not exceeding the system's maximum capacity. This formulation ensures that the energy storage operation respects the system's initial condition while preventing overcharging in later shifts. Constraints (16) define the range of values for the decision variables x_{jetow} .

3.3. Complexity analysis

In this subsection, we analyze the complexity of the proposed mathematical model.

Theorem 1.

The RCPSP-HE-TOU-MS is strongly NP-hard.

Proof. The RCPSP-HE-TOU-MS model degenerates to the RCPSP, which is known to be strongly NP-hard (Blazewicz et al., 1983). By omitting factors such as multiple shift systems, activity partitioning, TOU tariffs and the distinction between GE and power grid, the problem simplifies to an optimization model that minimizes total resource costs, as shown in Eq. (17):

$$\min \sum_{k=1}^k C_k(a_k) \quad (17)$$

The cost function $C_k(a_k)$ represents a discrete, non-decreasing function dependent on the amount of resource k acquired. In this simplified model, the RCPSP-HE-TOU-MS is reformulated as a Resource Investment Problem (RIP) or Resource Availability Cost Problem (RACP). Unlike the traditional RCPSP, which focuses on minimizing the project duration under resource constraints, the RIP/RACP aims to minimize the total costs within project duration limits, assuming unlimited resources. Consequently, the RIP/RACP is same kind of dual problem of

the RCPSP. Solving the RIP/RACP is classified as NP-hard (Möhring, 1984). ■

3.4. Exact solutions

We introduce a weighted-sum method designed to compute exact solutions for small-scale instances of the RCPSP-HE-TOU-MS. This method serves as a benchmark for evaluating PD-MODRL algorithms. The procedure of the weighted-sum method is outlined as follows:

Step 1: Assign an initial weight w_1 to the first objective function and $w_2 = 1 - w_1$ to the second objective function, where $w_1, w_2 \in [0, 1]$. Formulate a single-objective optimization problem by combining the two objectives using their respective weights. Use the Gurobi Optimizer to solve this optimization problem and record the solution. Add this solution to the solution set S , initializing $S = \emptyset$.

Step 2: Update the weights w_1 and w_2 by varying w_1 within the interval $[0, 1]$ in predefined steps (e.g., increments of 0.02 or smaller for finer resolution). For each pair of weights, repeat the optimization process in Step 1 to obtain a new solution. Add each newly obtained solution to the solution set S .

Step 3: Once all predefined weight combinations have been evaluated, examine the solution set S . Remove any dominated solutions to construct the Pareto frontier.

Step 4: Terminate the procedure. The remaining non-dominated solutions in the solution set S represent the Pareto frontier, providing a trade-off between the two objectives under different weight configurations.

4. PD-MODRL algorithm for the RCPSP-HE-TOU-MS

In Section 3.3, we demonstrated that the RCPSP-HE-TOU-MS is NP-hard. For large-scale project instances, exact algorithms cannot solve them in an acceptable time. While heuristic approaches like GA (Sprecher and Kolisch, 1996) offer potential solutions to this combinatorial optimization problem, their computational efficiency deteriorates significantly with increasing problem size. To address this limitation, we propose a reinforcement learning framework as a computationally efficient alternative. RL enables an agent to optimize long-term performance by interacting with the environment, receiving rewards, and adapting its policy to maximize cumulative returns. The process is illustrated in Fig. 2.

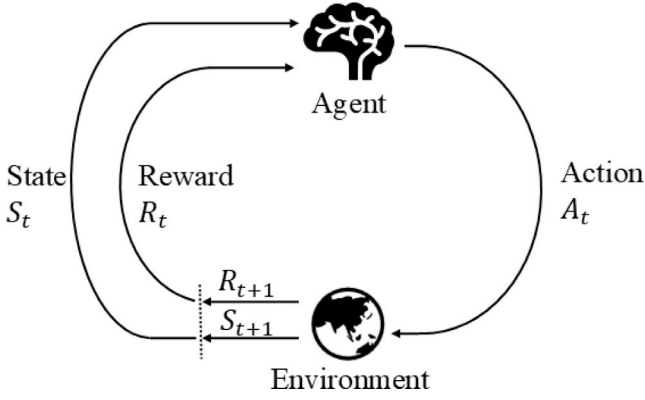


Fig. 2. Interaction process of reinforcement learning.

In this section, we present the PD-MODRL algorithm, which leverages a deep neural network to estimate the Q-values for each action (i.e., an executable activity segment) and subsequently selects actions to construct a complete scheduling plan. During the training process, the deep neural network takes both the state and the preferences as input. Incorporating preferences as input parameters enables the trained model to generate optimal policies that are tailored to any user-specified preferences during runtime. Since these preferences effectively guide the decision making within the policy framework, the approach is referred to as PD-MODRL. This section mainly includes the following content. First, the RCPSP-HE-TOU-MS is described as an MDP. Second, the composite state representation, actions and reward functions are defined. Finally, the structure of the neural network, the offline training framework and online solving framework of the PD-MODRL are developed.

4.1. Markov decision process of the RCPSP-HE-TOU-MS

The scheduling process of the RCPSP-HE-TOU-MS is modeled as a sequential decision-making framework. Decision points, indexed by $n \in 1, 2, \dots, N$, depend only on the previous state, with the exception of the project initiation.

At each decision point, multiple activity segments from different activities may be available for execution. The set of all such activity segments is termed the pending activity segments set, denoted by χ . In this study, only one activity segment from a specific activity is scheduled for execution at each decision point. Due to the splittable characteristic of activities in this study, executing multiple segments of different activities simultaneously would lead to an excessively large action space, thus increasing the training time and the complexity of the neural network. The detailed reason for this approach will be discussed in Section 4.3. In the defined MDP, if χ is non-empty at a decision point, the agent selects an activity segment for execution based on the current state s_t and evaluates whether new activities satisfying the constraints (precedence and resource constraints) can be added to χ . The project then transitions to the next state s_{t+1} , and the agent obtains a corresponding reward. This iterative process continues until the set χ becomes empty. At any given time, the project's next state depends solely on its current state, s_t , and the agent's decision, a_t , which is also known as the Markov property. A MDP models decision-making in dynamic environments. At each step, an agent observes the current state, selects an action, and transitions to a new state while receiving a reward. The process assumes the Markov property, meaning the next state depends only on the current state and action, not past history. The goal is to find an optimal strategy that maximizes long-term rewards. The primary components of the MPD framework are elaborated in the following subsections.

4.2. State

The state at decision point t is denoted as s_t , which incorporates several key sets: the set of completed activities (C_t), the set of completion times for scheduled activities (F_t) the set of completion times for activity segments (A_t), and the consumption of electricity, including both green electricity r_t^{ge} and grid electricity r_t^{grid} for executed activities at time t . Thus, the state at time t is represented as $s_t = \{C_t, F_t, A_t, R_t\}$, where $R_t = (r_t^{ge}, r_t^{grid})$. The initial state, s_0 , is defined by the scheduling problem's given information.

4.3. Action

At each decision point, the agent chooses a single activity segment to schedule. It is crucial to note that the action is a single activity segment, not a set of activity segments. Once an activity segment is scheduled, the MDP model transitions to the next stage, s_{t+1} , after the agent selects an activity segment from χ as the action a_t .

To reduce the complexity and computational time of neural network training caused by a large action space, we recommend executing one activity segment at a time. This strategy significantly reduces the action space, though it leads to a slight increase in the state space. For instance, consider two activities, j_1 and j_2 , which can be executed concurrently in state s_1 . Activity j_1 consists of 3 segments, and j_2 has 4 segments. The early start (ES) and latest finish (LF) times for j_1 are 0 and 6, respectively, providing 6 possible shifts. For j_2 , the ES is 0, and the LF is 8, offering 8 possible shifts. If both activities are executed simultaneously, the action space size is $\Lambda_1 = C_6^3 C_8^4 = 1400$. In contrast, executing one segment at a time reduces the action space. Assuming that j_1 is executed first, with its segments completed before those of j_2 , the action space at each state would be as follows: $\Lambda_1 = C_6^1 = 6$ at s_1 , $\Lambda_2 = C_5^1 = 5$ at s_2 , $\Lambda_3 = C_4^1 = 4$ at s_3 , $\Lambda_4 = C_8^1 = 8$ at s_4 , $\Lambda_5 = C_7^1 = 7$ at s_5 , $\Lambda_6 = C_6^1 = 6$ at s_6 , and $\Lambda_7 = C_5^1 = 5$ at s_7 . While the state space increases by 6, the action space is reduced by 1360.

Inspired by the findings in [55], where (Basaklar et al., 2022) observed that “the preference vectors have similar directional angles to the corresponding vectorized Q-values for a given state”, we integrate this insight into our approach. Specifically, we adopt their use of the cosine similarity between the preference vector and the vectorized Q-values within the Bellman optimality operator to guide the training process. This allows our method to better align the learned Q-values with the underlying preferences, enhancing the effectiveness of the optimization. To make the predicted Q-values closer to the target Q-values, we typically minimize the loss function every k steps. $L_k(\theta^{eval}) = \mathbb{E}_{(s_t, a_t, r_t, s_{t+1}, \Omega)} D[(y - Q(s_t, a_t, \Omega; \theta^{eval}))^2]$. Where θ^{eval} represents the parameters for evaluating the network. The experience replay buffer, denoted as D , stores transitions $(s_t, a_t, r_t, s_{t+1}, \Omega)$ for each time step. The target value y is defined as $y = r + \gamma Q(s_{t+1}, \sup(s_c(\Omega, Q(s_{t+1}, a_{t+1}, \Omega)) \cdot (\Omega^T Q(s_{t+1}, a_{t+1}, \Omega)); \theta^{target}))$, where θ^{target} represents the parameters of the target Q-network. Ω denotes the set of weights corresponding to each objective function. $S_c(\Omega, Q(s_{t+1}, a_{t+1}, \Omega))$ represents the cosine similarity between the preference vector and the Q-values. In the absence of this similarity term, the supremum operator selects the action that maximizes $\Omega^T Q(s_{t+1}, a_{t+1}, \Omega)$, potentially leading to a misalignment of the target Q-values, particularly when the objective scales differ significantly. Consider a multi-objective problem where the first objective is energy efficiency ($\in [0, 1]$) and the second is latency ($\in [1, 10]$). Suppose the preference vector $\Omega = \{0.9, 0.1\}$ prioritizes energy efficiency. There are two actions, a_1 and a_2 , with corresponding Q-values $Q_1 = \{0.9, 1\}$ and $Q_2 = \{0.1, 10\}$. Using the supremum operator, action a_2 is selected, as $\Omega^T Q_2 = 1.09$ exceeds $\Omega^T Q_1 = 0.91$. However, because the two objectives have vastly different magnitudes, this selection tends to favor latency, leading the training process in the wrong direction. In contrast, our proposed cosine similarity term aligns the preference vector with the Q-values, ensuring that the chosen actions maximize the target value. The modified operator

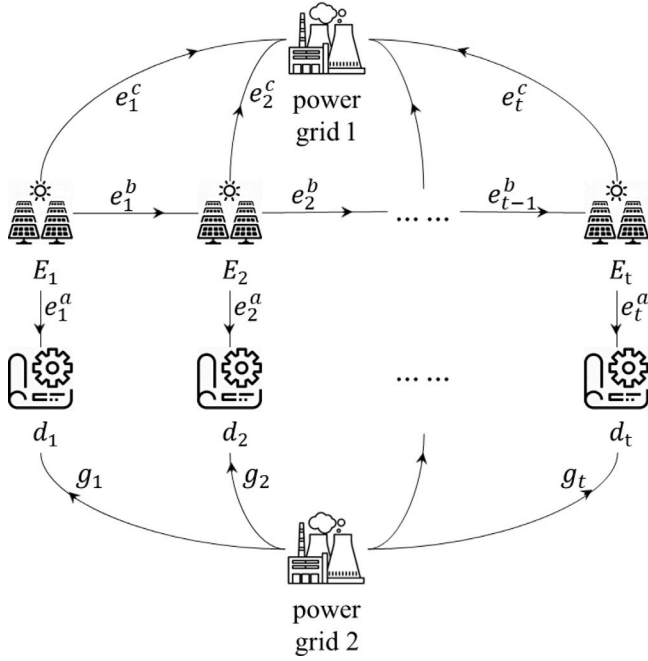


Fig. 3. Energy consumption network diagram.

now selects action a_1 , as $S_c \cdot \Omega^T Q_1 = 0.75 \times 0.91 = 0.68$, which is greater than $S_c \cdot \Omega^T Q_2 = 0.12 \times 1.09 = 0.13$. This adjustment ensures that actions with misaligned preferences and Q-values are less favored, as the similarity term $S_c(\Omega, Q(s_{t+1}, a_{t+1}, \Omega))$ assigns lower values in such cases.

4.4. Reward

After executing the chosen action, the agent transitions to the next state s_{t+1} and receives a reward r_t . The reward r_t evaluates action quality, directing policy optimization toward optimal decisions. The objective of RL is to train the agent to maximize the accumulated reward. In this study, the reward function is defined in Eq. (18).

$$r_t = TC_t - TC_{t+1} \quad (18)$$

Let TC_t represent the total cost of scheduled activities at decision point t . The energy demand from the initial time to t is determined by state s_t , allowing the exact computation of TC_t using an exact algorithm. Fig. 3 illustrates the network structure for calculating TC_t , which includes four types of nodes: GE capacity (E_t), demand (d_t), external power grid supply (Power Grid 2) and electricity sold (Power Grid 1). Energy demand at any given time can be met either by the external power grid (with no upper limit) or by the GE (with an upper limit). The flow balance in the network leads to the following equation: the total electricity purchased from the external power grid plus the GE equals the total demand plus the electricity sold. This problem is thus reformulated as a minimum cost flow problem on a directed network graph, where the GE and the external power grid nodes are sources, and the demand and the electricity sold nodes act as sinks. The network simplex method provides an efficient solution to this problem.

4.5. The architecture of the neural network

A neural network serves as the Q-function approximator in PD-MODRL. This section introduces a novel architecture designed to accommodate the proposed states. By incorporating spatial pyramid pooling (SPP) into the convolutional neural network (CNN), the model improves its ability to handle inputs of varying sizes. This enhancement

boosts the generalization of trained strategies across problem instances of different scales, thereby increasing the model's practical flexibility.

Before employing neural networks, it is essential to preprocess the inputs, which primarily consist of states, actions and preferences. The composition and generation mechanisms of states and actions are detailed in Sections 4.2 and 4.3. The generation process of the preference values involves two stages. In the first stage, preference vectors (w_1, w_2) are randomly sampled from a uniform distribution, where $w_1, w_2 \in [0, 1]$ and satisfy the condition $w_1 + w_2 = 1$. In the second stage, the generation of preference values is systematically adjusted rather than being random. The values of w_1 and w_2 are varied: w_1 starts at 1 and gradually decreases, while $w_2 = 1 - w_1$, until w_1 reaches 0. Fig. 4 shows the conversion of the sets C_t and F_t into vectors for state s_t (depicted in blue). Completed activities are indicated by 1 in the first row of the matrix, while their respective completion times are represented in the second row. Ω (the red region in the figure) denotes the weight set associated with each objective function. Notably, in this study, the weights are positioned as centrally as possible within the matrix. This design effectively mitigates boundary effects, improves the network's feature perception ability, and simultaneously enhances model training efficiency. The matrix in the figure includes the set O_t (shown in gray), which indicates the completion times of activity segments by marking positions with 1. The set R_t (shown in green) updates the matrix to capture energy consumption from both green and grid electricity for each completed activity at a given time. The final input data to the neural network are determined by the action a_t and the sets A_t and O_t . Given that F_t and R_t , which represent activity completion times and energy usage, contain large numerical values, normalization of these sets is required. Additionally, the matrix height adapts dynamically to changes in activity duration. The integration of SPP technology allows the neural network to adjust the matrix height dynamically without the need for retraining, improving the model's flexibility and scalability.

In the PD-MODRL algorithm, state-action pairs are fed into a neural network to estimate their Q-values. To handle complex states and actions, a modified CNN architecture, illustrated in Fig. 5, was developed. The preprocessed data first passes through a convolutional layer with multiple filters, generating a series of feature maps. These maps are then processed by an SPP layer, which partitions the maps into regions and applies pooling operations to produce a fixed-length feature representation. The resulting feature vectors are then forwarded to a fully connected layer. This layer splits into two parallel branches: the first computes the state value, while the second calculates the advantage value. Finally, the action value is derived by combining the state and advantage values.

4.6. The offline training framework of the PD-MODRL

The PD-MODRL algorithm enhances the traditional Deep Q-network (DQN) by integrating several advanced techniques: double DQN, dueling network architecture, multi-step methods and prioritized experience replay. Unlike conventional DQN, where the target network both evaluates and selects state-action pairs, PD-MODRL separates these functions. The evaluation network selects the state-action pair, while the target network assesses its Q value. Additionally, PD-MODRL employs a dueling network architecture, which divides the network into two branches before the output layer: one for estimating the state value function $V(s, \Omega)$ and the other for the advantage function $A(s, a, \Omega)$. These components are combined to compute the Q value, effectively reducing the risk of Q value overestimation. The computation is outlined in Eq. (19).

$$Q(s, a, \Omega; \theta^A, \theta^V) = V(s, \Omega; \theta, \theta^V) + \left(A(s, a, \Omega; \theta, \theta^A) - \frac{1}{A} \right) \sum_{a'} A(s, a', \Omega; \theta, \theta^A) \quad (19)$$

In the model, θ denotes the shared parameters of the value network, a' represents the set of feasible actions in the action space A , and θ^A and

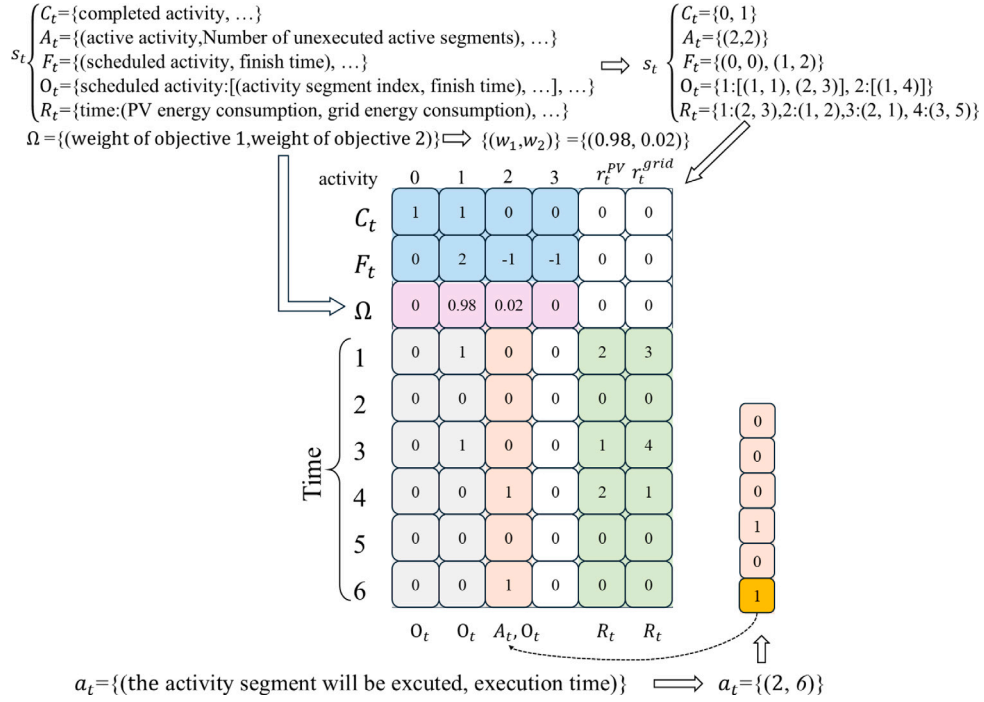


Fig. 4. An illustrative example of input composition.

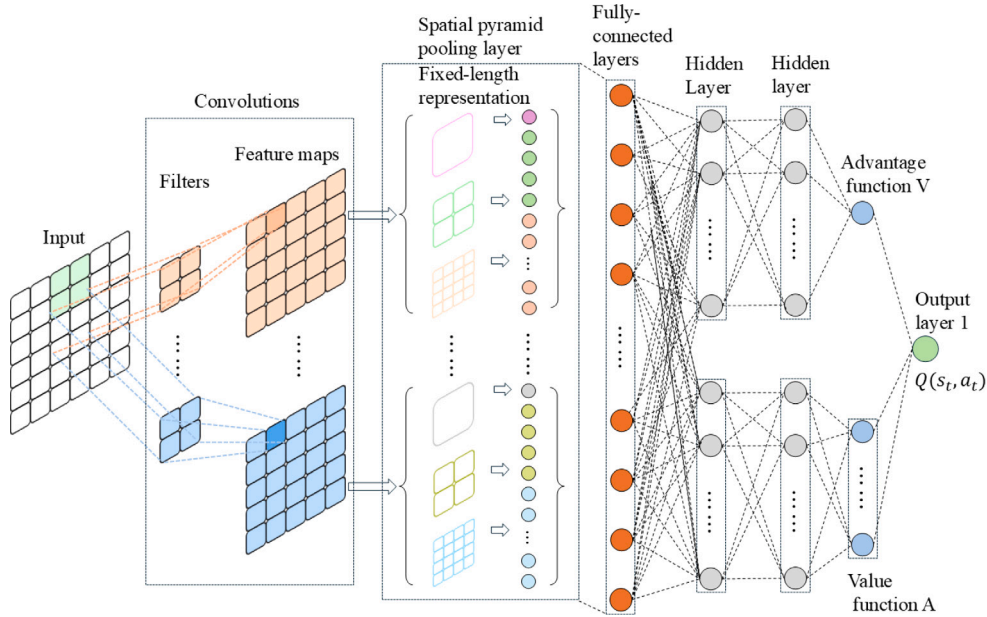


Fig. 5. Network architecture.

θ^V correspond to the parameters of the advantage and state-value functions, respectively. A multi-step approach is employed, distinguishing it from the standard DQN. Rather than relying on a one-step return based on the current reward and the Q value of the next state, the multi-step method incorporates rewards from several future steps, thus utilizing delayed reward signals more effectively. This improves the sample efficiency and accelerates the learning process. The Bellman equation, as shown in Eq. (20), defines lr as the learning rate and m as the number of steps. In Q-learning, the lr controls the step size of updates, balancing prior estimates with new information. Additionally, model

convergence is expedited through prioritized sampling from the experience replay pool, in contrast to traditional uniform random sampling. This technique assigns varying priorities to experiences, ensuring that those more critical for policy improvement are resampled with higher frequency. For further details, please refer to the relevant literature [59].

$$Q(s_{t+1}, a_{t+1}, \Omega) = (1 - lr)Q(s_t, a_t, \Omega) + lr \left(\sum_{b=0}^{m-1} r_{t+b} + \max_{a_{t+m}} Q(s_{t+m}, a_{t+m}, \Omega) \right) \quad (20)$$

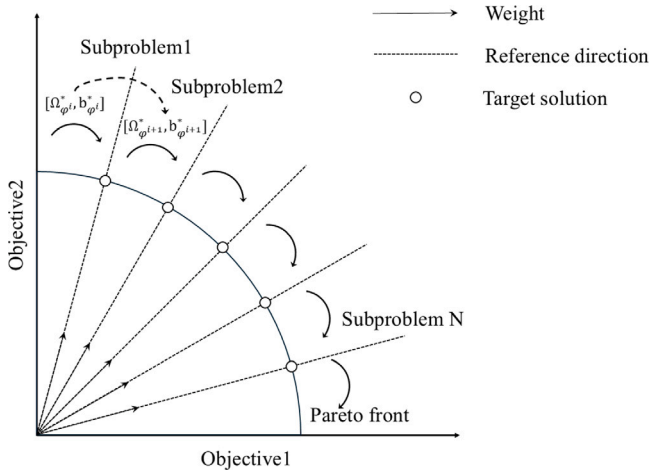


Fig. 6. Illustration of the parameter-transfer strategy.

The training framework is outlined in Algorithm 1. To improve the generalization of PD-MODRL, a random instance is sampled at the start of each episode. An ϵ -greedy policy is employed for action selection, balancing exploration and exploitation. ϵ decays exponentially, as shown in Eq. (21).

$$\epsilon = \epsilon_{\min} + (\epsilon_{\max} - \epsilon_{\min}) e^{-n_{\text{step}} \cdot d_r} \quad (21)$$

Here, n_{step} represents the accumulated number of steps for the agent to interact with the environment, d_r is the decay rate of ϵ , while $\epsilon_{\min}$ and $\epsilon_{\max}$ denote the minimum and maximum ϵ , respectively.

To obtain Pareto front solutions, we assign different weights to the objective functions. First, a set of uniformly distributed weight vectors $\Omega = \{\varphi^1, \varphi^2, \dots, \varphi^N\}$ is provided. For instance, in a bi-objective problem, these vectors could include (1, 0), (0.08, 0.92), ..., (0, 1). Specifically, $\varphi^j = (\varphi_1^j, \dots, \varphi_M^j)$, where M denotes the number of objectives. In other words, to obtain Pareto front solutions, we train N similar subproblems using Algorithm 1. The objective function for solving these subproblems is defined as follows:

$$\text{minimize } g^{\Omega^j}(x | \varphi_i^j) = \sum_{i=1}^M \varphi_i^j f_i(x) \quad (22)$$

When the number of subproblems is small, Algorithm 1 can be used to train each subproblem sequentially. However, when the number of subproblems increases, training them individually becomes time-consuming. To address this issue, this paper adopts a Neighborhood-based parameter-transfer strategy, as proposed by Li et al. (2020), to reduce the model's training time. Based on Eq. (22), it can be observed that neighboring subproblems often have nearly identical optimal solutions, as their weight vectors are adjacent (Zhang and Li, 2007). Therefore, solving a subproblem can benefit from leveraging the solutions of its neighboring subproblems.

Specifically, we first solve an initial subproblem using Algorithm 1, where the parameters of the neural network model are represented as $[\Omega_{\varphi^i}^*, b_{\varphi^i}^*]$. The optimized parameters of the neural network model are denoted as $[\Omega_{\varphi^i}^*, b_{\varphi^i}^*]$. Next, we sequentially train subproblems $i+1, i+2, \dots, i+N-1$, using the trained initial subproblem as the starting point. In brief, the network parameters are transferred sequentially from one subproblem to the next, as illustrated in Fig. 6. The neighborhood-based parameter transfer strategy significantly reduces the training time of PD-MODRL, thereby enhancing the efficiency of obtaining N models.

4.7. The online solving framework of the trained PD-MODRL

Upon completing the offline training, the agent shifts from the ϵ -greedy policy used during training to an online policy for problem

Table 4

Problem parameter settings for generating the RCPSP-HE-TOU-MS instances.

Parameters	Parameter setting
D	$\text{Round}(0.6 \times \sum_{j=1}^J d_j)$
$P_{\max}$	$U(10, 30)$
P_{to}	$U(0, 1.3P_{\max})$
Q	$\text{Round}(0.6P_{\max})$
Z	$\text{Round}(0.7 \sum_{j=1}^J \mu \cdot P_j \cdot d_j)$
C_s	0.5
$C_{f\omega}$	$0.725(\text{peak period})/0.385(\text{valley period})$
C_g	0.1
C_e	0.2
μ	0.68

solving. In this phase, it evaluates the Q-values of available actions and selects the one with the highest Q-value at each step, striving to approximate an optimal solution. The online solution process is outlined in Algorithm 2. Algorithm 2 is applied sequentially to each of the N subproblem models, which have been trained previously. Given that the N subproblems are inherently parallel, they can be executed simultaneously, which accelerates the overall process and allows for the rapid acquisition of Pareto frontier solutions.

5. Experimental results and analysis

To assess the effectiveness of the proposed algorithm, we first independently applied PD-MODRL and the Gurobi Optimizer 10.0.1 solver to solve small-scale instances. For large-scale instances, this study compares PD-MODRL with NSGA-II, a widely used algorithm for large-scale optimization, and with the classical RL approach ResNet-DQN, to provide a comprehensive performance evaluation.

Extensive computational experiments were performed on a server equipped with an Intel(R) Core (TM) i9-13900HX CPU (13th Gen, 2.20 GHz) and 64 GB RAM. To ensure fairness, all experiments were executed on a single thread. The algorithms were implemented in Python, and neural networks were constructed using TensorFlow 2.6.0.

5.1. Data generation

Due to the absence of standard instances for the RCPSP-HE-TOU-MS and the challenge of obtaining real-world data from construction projects, new instances were created by extending the standard PSPLIB benchmarks (Sprecher and Kolisch, 1996) to evaluate the performance of PD-MODRL across various problem sizes. These extended datasets preserve the number of project activities and precedence found in the classic RCPSP test cases. The peak power $P_{\max}$ is set within the range defined by the original RCPSP parameters. According to the Chinese Academy of Engineering, the unit standard carbon emission coefficient for coal-fired power generation is 0.68, hence $\mu=0.68$. The relevant problem parameters are presented in Table 4.

5.2. Validating the proposed algorithm

This section compares the performance of the proposed algorithm with that of exact algorithms in solving small-scale problems. The exact solution is only able to solve project instances with up to 20 activities, and the computational time is significantly long. To assess the effectiveness of PD-MODRL, we randomly selected five test cases from the J10.mm and J20.mm instance sets. The performance of the PD-MODRL algorithm is compared with the exact solution for the small-scale project instances. In the comparative analysis, the solution with the smallest value of the first objective in the Pareto solution set of the two algorithms is selected for comparison first, followed by the solution with the smallest value of the second objective. This method effectively

Algorithm 1 The training procedure of the PD-MODRL with proportional prioritization

```

1: Input: the number of episodes  $EP$ , the number of steps in the multi-step update
    $Mstep$ , the replay size  $N$ , the number of training interval steps  $interval$ , the batch
   size  $H$ ; exponents  $\alpha$  and  $\beta$ , the interval steps for replacing target network weights
    $rep$ , the learning rate  $lr$ ,  $\epsilon_{max}$  and  $\epsilon_{min}$ , the set of weights for the objective function,
   denoted as  $\Omega$ 
2: Output: the weights of the evaluation network  $\theta^{eval}$ 
3: Initialize: the weights of evaluation network  $\theta^{eval}$  (random), the weights of the
   target network  $\theta^{targ} \leftarrow \theta^{eval}$ , the replay memory  $H = \emptyset$ ,  $p_1 = 1$ , the number of
   steps for the agent to interact with the environment  $n_{steps} = 0$ 
4: for  $episode = 1$  to  $EP$  do
5:   Sample an instance
6:   Reset the environment
7:    $t = 1, s_t = s_1, done = False, \epsilon = \epsilon_{max}, mstep = 0, exp = \emptyset$ 
8:   while not done do
9:      $mstep = mstep + 1$ 
10:     $n_{steps} = n_{steps} + 1$ 
11:    Get the action space  $\tilde{A}_t$ 
12:    if random number  $< 1 - \epsilon$  do
13:       $a_t = \operatorname{argmax}_{a_t \in \tilde{A}_t} Q(s_t, a_t, \Omega | \theta^{eval})$ 
14:    else
15:       $a_t =$  randomly choose  $\tilde{A}_t$ 
16:    end if
17:    Update  $\epsilon$  with exponential decay
18:    Execute  $a_t$ 
19:    Receive the reward  $r_t$ , observe the new state  $s_{t+1}$ , update  $done$ 
20:    Store transition  $(s_t, a_t, r_t, \Omega)$  in  $exp$ 
21:    if  $mstep = Mstep$  or  $done$  do
22:      Store transition  $(s_{t+1})$  in  $exp$ 
23:       $mstep = 0$ 
24:    end if
25:    if the length of  $H < N$  do
26:      Store  $exp$  in the end of  $H$  with maximal priority
27:       $p_t = \max_{i < t} p_i$ 
28:    else
29:      Delete the first experience of  $H$ 
30:      Store  $exp$  in the end of  $H$  with maximal priority
31:       $p_t = \max_{i < t} p_i$ 
32:    end if
33:     $s_t = s_{t+1}, t = t + 1$ 
34:    if all activities are scheduled do
35:       $done = True$ 
36:    end if
37:    if  $n_{steps} \bmod interval = 0$  do
38:      for  $j=1$  to  $H$  do
39:        Sample transition  $j \sim P(j) = p_j^\alpha / \sum_i p_i^\alpha$ 
40:        Compute importance-sampling weight
41:         $\omega_j = (N \cdot P(j))^{-\beta} / \max_i \omega_i$ 
42:        Compute TD-error  $\delta_j$ 
43:        Update transition priority  $p_j \leftarrow |\delta_j|$ 
44:      end for
45:      Train PD-MODRL and update  $\theta^{eval}$ 
46:      Update  $lr$  with exponential decay
47:    end if
48:    if  $n_{steps} \bmod rep = 0$  do
49:       $\theta^{targ} \leftarrow \theta^{eval}$ 
50:    end if
51:  end while
52: end for

```

evaluates the performance of the designed algorithm by optimizing the objectives. This study quantifies the performance disparity between

Gurobi and PD-MODRL using the Gap metric, which is formally defined by the following mathematical expression:

Algorithm 2 The online solving framework of the trained PD-MODRL with proportional prioritization

```

1: Input:  $\theta^{eval}, \Omega$ 
2: Output: activity list  $AL = \emptyset$ , start time  $ST = \emptyset$ ,  $TC = 0$ 
3: Initialize:  $t = 0$ ,  $s_t = s_0$ ,  $done = False$ ,  $AL = \emptyset$ ,  $ST = \emptyset$ , current time  $\zeta = 0$ 
4: while not done do
5:   Get the action set  $\tilde{A}_t$ 
6:   Evaluate all actions in  $\tilde{A}_t$ 
7:   Choose the action with the maximum Q value
      $a_t = \underset{a_t \in \tilde{A}_t}{\operatorname{argmax}} Q(s_t, a_t, \Omega \mid \theta^{eval})$ 
8:   Update AL and ST according to  $a_t$  and  $\zeta$ 
9:    $s_t = s_{t+1}$ ,  $t = t + 1$ , update current time to  $\zeta + 1$ 
10:  if all activities are scheduled do
11:     $done = True$ 
12:  end if
13: end while
14: Calculate TC

```

Table 5

Comparison of solution results for minimizing objective 1.

Number of activities	Number of instances	Energy cost		Gap (%)	Carbon emission		Gap (%)
		Gurobi	PD-MODRL		Gurobi	PD-MODRL	
10	1	41.99	42.49	-1.18	87.04	45.56	91.04
	2	70.14	67.02	4.66	69.36	56.44	22.89
	3	32.25	32.32	-0.22	85.68	38.08	125.00
	4	33.10	28.52	16.06	51.68	59.44	-13.06
	5	34.68	29.95	15.79	70.72	30.60	131.11
Average	–	42.43	40.06	7.02	72.90	46.02	71.40
20	1	498.37	517.40	-3.68	471.92	422.28	11.76
	2	449.78	456.23	-1.41	705.16	389.64	80.98
	3	414.90	439.90	-5.68	488.92	369.92	32.17
	4	519.13	501.58	3.50	635.80	406.64	56.35
	5	528.69	570.55	-7.34	550.12	512.12	7.42
Average	–	482.17	497.13	-2.92	570.38	420.12	37.74

Table 6

Comparison of solution results for minimizing objective 2.

Number of activities	Number of instances	Energy cost		Gap (%)	Carbon emission		Gap (%)
		Gurobi	PD-MODRL		Gurobi	PD-MODRL	
10	1	52.10	46.97	10.92	0.00	4.08	-100
	2	73.54	70.10	4.91	40.12	29.92	34
	3	41.70	34.02	22.57	0.00	0.68	-100
	4	39.10	32.86	18.99	0.00	2.04	-100
	5	42.90	32.95	30.20	0.00	3.40	-100
Average	–	49.87	43.38	17.52	8.02	8.02	-73.18
20	1	526.79	527.73	-0.18	223.04	263.16	-15.25
	2	493.71	460.39	7.24	317.56	278.12	14.18
	3	425.80	447.8	-4.91	395.08	369.24	7.00
	4	553.34	518.93	6.63	341.36	305.32	11.80
	5	532.02	576.17	-7.66	515.44	493.68	-4.22
Average	–	506.33	506.20	0.22	358.50	341.90	2.70

$$\text{Gap}(\%) = \frac{R_{\text{Gurobi}} - R_{\text{PD-MODRL}}}{R_{\text{PD-MODRL}}} \quad (23)$$

In the formula, R_{Gurobi} represents the result obtained from the Gurobi optimizer, while $R_{\text{PD-MODRL}}$ represents the outcome derived from the PD-MODRL algorithm. Both R_{Gurobi} and $R_{\text{PD-MODRL}}$ encompass energy costs and carbon emissions.

Table 5 presents the comparative results for minimizing the first objective. The PD-MODRL algorithm outperforms Gurobi in both energy consumption and carbon emissions. Specifically, for instances with 10

activities, Gurobi increases carbon emissions by an average of 71.40%, whereas for 20 activities, the increase is 37.74%. Regarding energy costs, the difference is less pronounced: Gurobi exceeds PD-MODRL by an average of 7.02% for 10 activities and reduces costs by 2.92% for 20 activities.

Table 6 presents the results for minimizing the second objective. In this case, Gurobi results in an average energy cost that is 17.52% higher for 10 activity instances and 0.22% higher for 20 activity instances compared to PD-MODRL. For 10 activity instances, Gurobi achieves

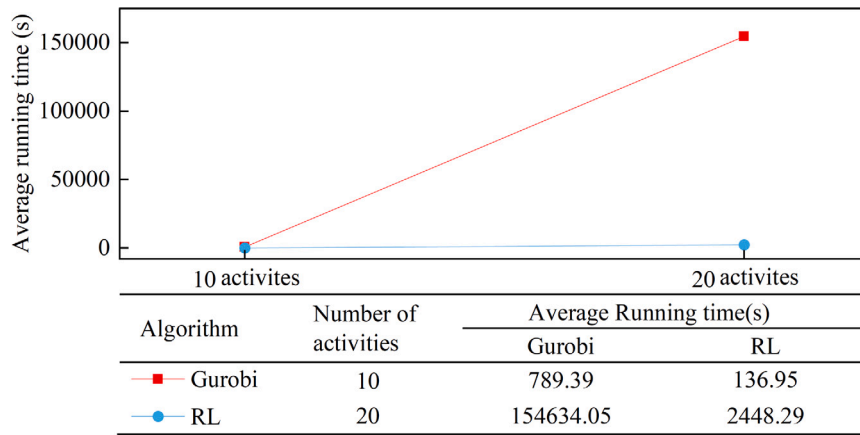


Fig. 7. The average solution time of the two algorithms for the test cases.

zero carbon emissions, which is expected since the second objective is prioritized for optimization. This result is primarily due to the extensive use of renewable energy during optimization. Although PD-MODRL does not always achieve zero emissions, the absolute difference from Gurobi's results remains small. However, with 20 activities, Gurobi shows a slightly larger average carbon emission gap compared to PD-MODRL.

Overall, the PD-MODRL algorithm performs well in reducing carbon emissions and optimizing energy consumption, with its scheduling strategy minimizing energy costs while simultaneously reducing carbon emissions.

Fig. 7 illustrates that as the number of activities increases, Gurobi's solution time grows significantly, highlighting its limited scalability and computational efficiency. In contrast, although the runtime of the PD-MODRL algorithm increases, its growth rate is much slower than that of Gurobi, highlighting its advantage in solving large-scale problems. This result suggests that the PD-MODRL algorithm maintains stable and efficient performance in larger scheduling problems, offering superior scalability and potential for broader application.

5.3. Results of large instances

To evaluate the performance of the proposed PD-MODRL algorithm on large-scale instances, this study tests it on J30.mm, J60.mm, and J90.mm datasets, which consist of 30, 60 and 90 activities, respectively. To expedite the evaluation, each algorithm is applied to a randomly selected set of 100 project instances. Exact algorithms are excluded from this analysis because solving most instances within a reasonable time is infeasible.

Hyperparameter tuning plays a critical role in determining the performance of Q-networks in RL. Due to the high-dimensional search space, a random search strategy was adopted to identify effective configurations (Bergstra and Bengio, 2012). To evaluate the effects of lr and d_r , a benchmark was constructed using adjusted J10, J20, and J30 instances, with five cases randomly selected from each set under a fixed seed to ensure reproducibility. In Fig. 8, the x -axis represents training iterations, and the y -axis shows the weighted objective value. Fig. 8 shows that extreme lr values degrade performance, with 0.005 identified as the optimal setting. Fig. 8(b) demonstrates that a d_r of 1/5000 provides a balance between computational efficiency and solution stability. NSGA-II is run for 1000 and 3000 iterations, and the results are reported accordingly.

The dominance ratio method, introduced by Zitzler and Thiele (1999), is employed to assess computational results. Specifically, it calculates the proportion of individuals on the Pareto front of one algorithm that are dominated by individuals on the Pareto front of

another algorithm. For two given Pareto fronts, A and B , the dominance ratio $C(A, B)$ represents the percentage of individuals in B dominated by one or more individuals in A . This ratio is calculated as follows:

$$C(A, B) = \frac{|\{b \in B \mid \exists a \in A : b < a\}|}{|B|} \quad (24)$$

The smaller the value of $C(A, B)$, the better the quality of B (see Table 7).

As shown in Table 7, the PD-MODRL algorithm exhibits clear superiority across all benchmark tasks. Its C -values at 1000 and 3000 iterations are approximately 1/6 of those achieved by NSGA-II, indicating a substantial enhancement in solution quality. Compared with ResNet-DQN, the C -value is further reduced by approximately 20%, reaffirming the superior solution quality of PD-MODRL. In terms of computational efficiency, PD-MODRL reduces runtime by about 89.76% relative to NSGA-II, demonstrating remarkable efficiency, as reported in Table 8. Although its runtime is roughly 7.98% longer than that of ResNet-DQN, PD-MODRL achieves an optimal trade-off between solution quality and computational cost.

5.4. Sensitivity analysis of key parameters

Tables 9 and Fig. 9 illustrate the responses of TC and CE to variations in key parameters. Specifically, a single instance is randomly selected from the J30 dataset, and the price of electricity, energy storage capacity and peak power load are individually adjusted by $\pm 20\%$ and $\pm 30\%$ to compute the Pareto front under each scenario. Subsequently, two extreme points on each Pareto front are identified: M1 (minimal TC, maximal CE) and M2 (maximal TC, minimal CE). The variations of TC and CE before and after parameter perturbations are then quantified to evaluate the system's sensitivity to parameter changes.

The sensitivity analysis results in Table 9 and Fig. 9 indicate that parameter variations exert markedly different impacts on TC and CE. Electricity price is the most influential factor, causing substantial TC fluctuations: a 20% and 30% increase in price raises TC at the minimum-cost point M1 by 17.2% and 27.31%, respectively, while a 20% and 30% decrease reduces TC by 19.13% and 27.21%. In contrast, CE changes are minor, suggesting that electricity price primarily affects system performance through economic constraints rather than energy mix adjustments. Changes in energy storage capacity have limited impact on TC, below 1.1%, and moderate effects on CE, up to 6.9%, indicating strong system robustness and the potential for storage expansion to enhance energy dispatch and renewable energy utilization. Peak power load has negligible influence on TC, within $\pm 2\%$, but affects CE more significantly: an increase raises CE by approximately 10%, whereas

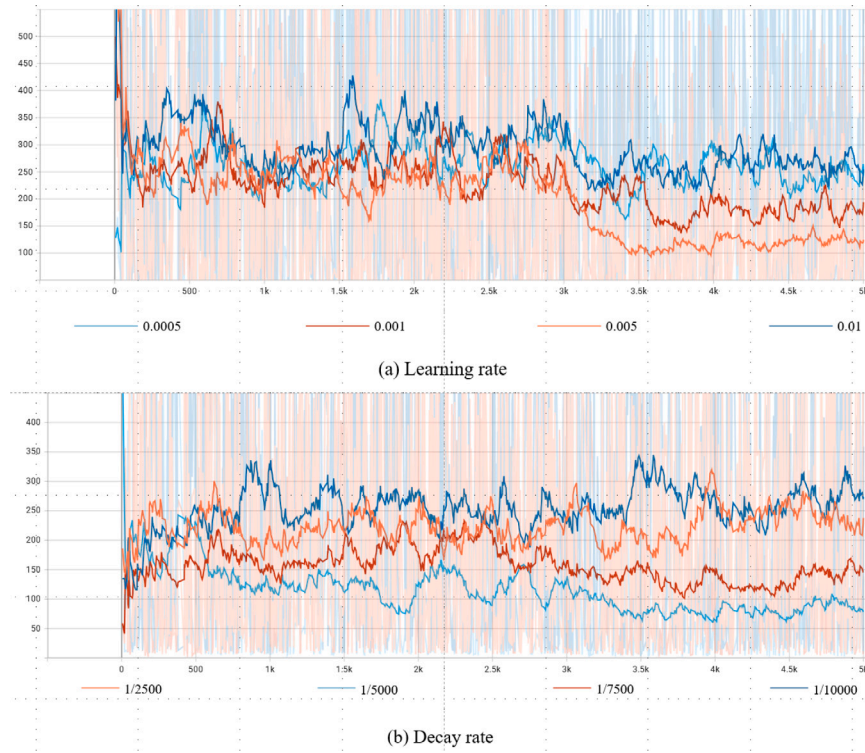


Fig. 8. Verification results of hyperparameter.

Table 7
Comparison of *C*-values between PD-MODRLS, NSGA-II and ResNet-DQN.

Item	1000 iterations		3000 iterations		<i>C</i> (A3, A1)	<i>C</i> (A1, A3)
	<i>C</i> (A2, A1)	<i>C</i> (A1, A2)	<i>C</i> (A2, A1)	<i>C</i> (A1, A2)		
J30	0.091	0.849	0.089	0.852	0.199	0.244
J60	0.069	0.906	0.115	0.847	0.260	0.3789
J90	0.265	0.735	0.233	0.767	0.344	0.364

Note: A1, A2 and A3 correspond to the PD-MODRL NSGA-II and ResNet-DQN algorithms, respectively.

Table 8
Comparison of run times between PD-MODRLS, NSGA-II and ResNet-DQN (Unit of time: second).

A1	1000 iterations	3000 iterations	A3
	A2	A2	
1240.21	68785.78	206809.45	1062.33
5686.32	83996.92	215001.77	6421.40
16473.98	83561.82	60510.01	12761.51

Note: A1, A2 and A3 correspond to the PD-MODRL, NSGA-II and ResNet-DQN algorithms, respectively.

a 30% reduction lowers it by 13.9%, reflecting indirect effects on emissions through altered energy usage patterns.

The sensitivity analysis offers several managerial insights for practitioners seeking to enhance energy-cost efficiency and emission performance in project scheduling. First, the substantial responsiveness of TC to electricity price changes underscores the necessity for managers to integrate dynamic pricing forecasts into operational planning. Strategies such as shifting noncritical activities to low-price periods or securing long-term electricity contracts can effectively mitigate financial risks. Second, the limited impact of storage capacity variations on TC, but its moderate influence on CE, suggests that investment in energy storage should be guided primarily by environmental objectives

rather than cost considerations. Managers aiming to reduce emissions may therefore prioritize moderate storage expansion to improve renewable energy integration without incurring significant additional costs. Finally, although peak power load adjustments exert minimal effects on TC, their pronounced influence on CE highlights the importance of peak-load management from an environmental standpoint. Implementing demand-response strategies, smoothing load profiles, or redesigning activity sequences to avoid emission-intensive peak periods can substantially reduce carbon impacts. Collectively, these findings provide actionable guidance for balancing cost and sustainability objectives under parameter uncertainty.

6. Case study

This section presents a case study to analyze the impact of hybrid energy supply and TOU tariffs on the total cost and carbon emissions. We explore the utilization of various energy types and associated costs across different periods, along with their distribution characteristics. The project concerns the manufacture of a customized piece of furniture with a fixed delivery deadline. This type of make-to-order production is representative of many engineer-to-order projects, where customer specifications require tailored processing and strict adherence to due dates. Several critical components require high-energy-consuming equipment, such as CNC machining centers and heat treatment furnaces, which dominate the project's electricity demand.

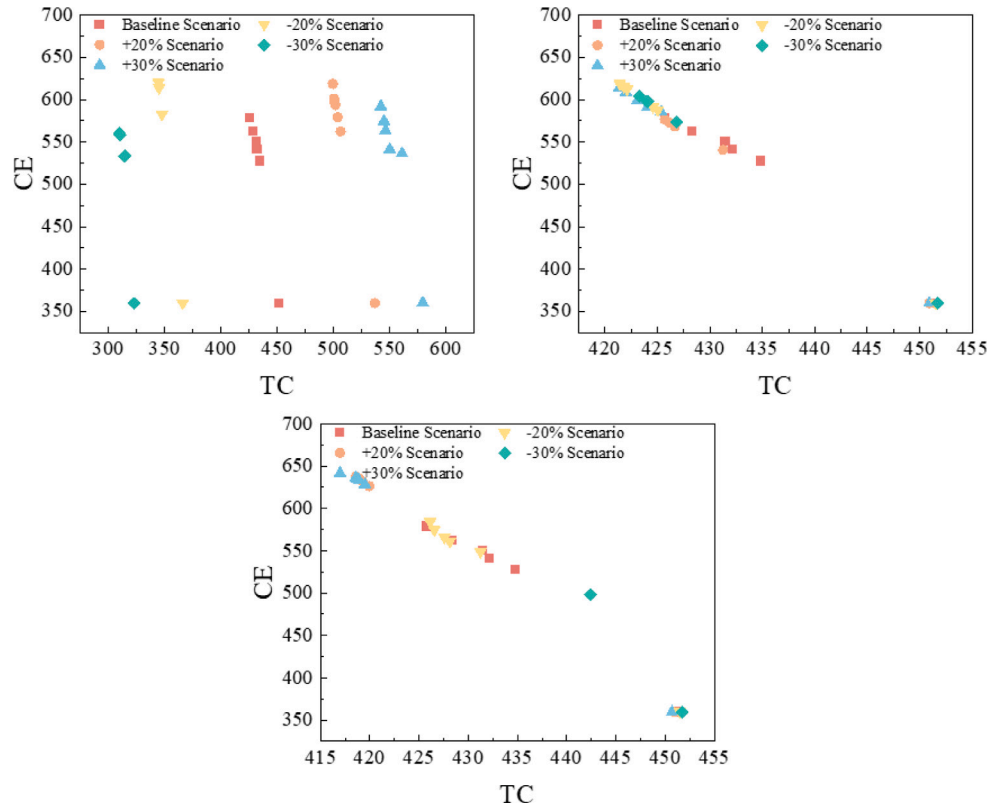


Fig. 9. Pareto frontier when key parameters change.

Table 9

TC and CE sensitivity to key parameter variations.

Parameter		+20%		+30%		-20%		-30%	
		TC	CE	TC	CE	TC	CE	TC	CE
Price of electricity	M1	17.27%	6.93%	27.31%	2.35%	-19.13%	7.17%	-27.21%	-3.17%
	M2	18.93%	0.00%	28.40%	0.00%	-18.93%	0.00%	-28.40%	0.00%
Energy storage capacity	M1	-0.01%	-0.35%	-1.04%	6.11%	-1.02%	6.93%	-0.59%	4.47%
	M2	-0.07%	0.00%	-0.07%	0.00%	0.07%	0.00%	0.11%	0.00%
peak power load	M1	-1.68%	10.22%	-1.69%	9.99%	0.08%	0.94%	3.90%	-13.87%
	M2	-0.07%	0.00%	-0.11%	0.00%	0.07%	0.00%	0.11%	0.00%

A process route was developed to align with the company's capabilities and component design, launching the project to meet the order requirements. Key high-energy-consuming equipment is essential for project completion. Detailed information regarding the duration and power requirements for each activity is provided in Table A1 in Appendix. The contract due date is set at 34 days, which reflects a realistic production cycle for such furniture projects while ensuring comparability with benchmark instances. The peak power limit of $P_{max} = 14$, representing typical factory capacity constraints for similar projects. In the absence of actual cost data, we assume cost values based on the artificial data instances in Table 4, which are consistent with PSPLIB-based parameters and industry-reported carbon coefficients.

The total cost for the 34-day contract period (68 shifts) was 220.498, representing a 38.16% reduction from the cost of 356.556 incurred without GE and TOU tariffs. Carbon emissions decreased by 17.81%, from 350.84 without GE and TOU tariffs to 297.84 with them. Without TOU tariffs, the electricity price is based on the average of the peak and off-peak rates. The final schedule, shown in Fig. 10, illustrates

that the proposed algorithm effectively schedules activities during low-cost periods, though some high-cost periods were unavoidable due to strict contractual constraints.

Fig. 10 also shows the temporal fluctuations of cost components in a dual-energy system under TOU tariffs. The total cost follows a steady trend, while four other cost categories fluctuate intermittently. These fluctuations are mainly driven by the integration of GE and TOU tariffs. The intermittency of GE sometimes necessitates supplementary grid electricity. When GE meets the real-time demand and surplus power charges the storage system, the system generates revenue by selling stored electricity during peak grid price periods. In contrast, during low-price periods, the system prioritizes energy storage over sales, even when GE alone can meet the project's needs, thereby maximizing economic benefits. Consequently, both the grid electricity cost curve (red) and the revenue curve (blue) show periodic changes, reflecting the combined impact of PV variability and electricity price volatility. During high-price periods, excess GE is sold, while in low-price periods, GE is stored, and inexpensive grid electricity is purchased, optimizing

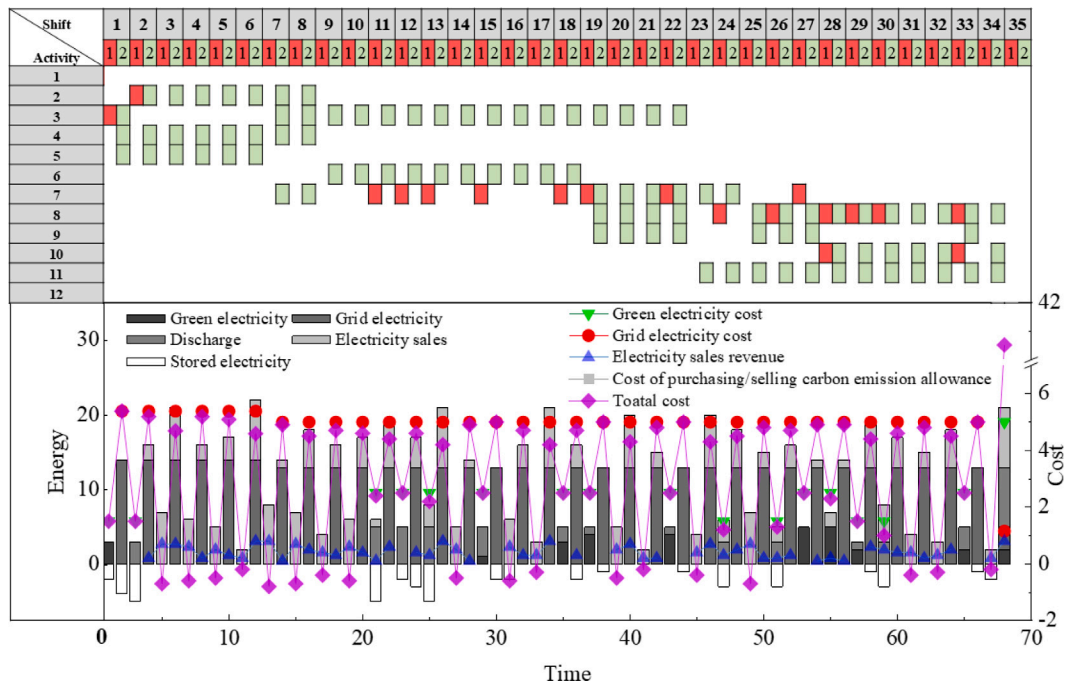


Fig. 10. Scheduling gantt chart and resource cost analysis.

both costs and revenues.

Overall, the results demonstrate that integrating renewable energy with TOU tariffs can effectively lower energy costs and carbon emissions. While strict contractual deadlines make some peak-hour usage unavoidable, the coordinated use of GE, storage and TOU tariffs still achieves significant benefits.

7. Conclusion and future research

Renewable energy and TOU tariffs have globally gained increasing attention as effective measures for energy conservation and carbon emission reduction. This paper addresses the integration of GE and TOU tariffs in the project scheduling problem, aiming to reduce carbon emissions and optimize energy costs by leveraging varying energy prices across different time periods. Despite extensive research on various RCPSP variants, studies incorporating GE and TOU tariffs in project scheduling are limited. This paper introduces a novel RCPSP variant, the RCPSP-HE-TOU-MS, which incorporates renewable energy, TOU tariffs and activity preemption within a multi-shift system. A mixed-integer linear programming model and an MDP model are developed for the RCPSP-HE-TOU-MS to minimize total costs and carbon emissions. Additionally, a CNN-based architecture is proposed to extract features from the problem and to facilitate decision making. PD-MODRL is utilized for end-to-end training, optimizing the model's performance. Experimental results demonstrate the effectiveness and scientific rigor of the proposed algorithm. The management implications of this research are as follows:

- (1) From an operational perspective, managers can achieve significant efficiency gains by adopting a synergistic approach that integrates multi-shift scheduling, renewable energy utilization, and TOU tariffs. The proposed model demonstrates the value of aligning high-energy-consuming activities with periods of low electricity tariffs and peak renewable generation. This integrated perspective allows firms to maximize the utilization of energy-intensive equipment while strictly adhering to project deadlines, thereby ensuring both economic efficiency and contractual

compliance.

- (2) In addition to operational optimization, the proposed framework enhances organizational resilience against the volatility of the global energy market. As policies increasingly shift towards dynamic pricing and renewable adoption, enterprises must possess the adaptability to navigate price fluctuations and supply intermittency. By leveraging the flexibility inherent in the RCPSP-HE-TOU-MS model, managers can proactively mitigate risks associated with peak power limitations and rising grid costs. This capability not only reduces exposure to market uncertainties but also transforms passive energy consumption into a competitive advantage within a sustainability-driven industry.
- (3) Ultimately, this research empowers decision-makers to navigate the complex trade-offs between profitability and environmental sustainability. The intelligent optimization algorithm provides diverse scheduling scenarios, enabling managers to visualize the marginal costs of carbon reduction. By quantifying the impact of energy decisions, firms can transition from compliance-driven actions to proactive carbon neutrality strategies, strengthening their corporate image and securing long-term strategic value.

The proposed optimization approach demonstrates clear advantages in theoretical analysis and model validation; however, some limitations remain. It assumes a predictable renewable energy supply and does not fully account for its inherent variability and uncertainty, potentially restricting real-world applicability. The method also depends on the availability of renewable energy facilities, making it more suitable for low-volume or workshop-based manufacturing enterprises with large rooftop installations, such as solar panels, and less effective for firms lacking such infrastructure.

Future research should address the uncertainty of renewable energy supply in project management. The integration of forecasting models for solar and wind generation can enhance scheduling algorithms and energy management strategies, thereby improving system robustness. In particular, studies should explore dynamic adjustment of resource allocation and scheduling under different shift systems and TOU tariffs,

Table A.1
Activity processing times, power and successor activities.

Activity ID	d_j	P_j	Successor activities
1	0	0	2/3/4/5
2	4	3	6/9
3	9	3	11
4	4	2	6/11
5	3	9	7/8
6	5	10	8
7	8	5	10
8	10	3	12
9	4	2	12
10	4	2	12
11	6	8	12
12	0	0	

aiming to maximize project profitability while minimizing energy costs. Such approaches are crucial for enhancing operational efficiency and decision-making flexibility in enterprises facing energy uncertainty.

CRedit authorship contribution statement

Liangwei Chen: Writing – original draft, Methodology, Data curation. **Jingwen Zhang:** Writing – review & editing, Supervision. **Zhi Chen:** Investigation, Formal analysis, Data curation. **Erik Demeulemeester:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

See [Table A.1](#)

Data availability

Data will be made available on request.

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